

PREDICTIVE BEATS: DUAL-ENGINE COLLABORATIVE FILTERING



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ABSTRACT

This research develops a dual-engine music recommendation system combining user-based collaborative filtering (UB-CF) and item-based collaborative filtering (IB-CF), applied to historical listening data via cosine similarity. Both engines surface Rock-dominant recommendations for the target user — UB-CF drawing from taste twin overlap and IB-CF anchoring on the seed track "Something So Strong" by Crowded House.

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DATASETS

1.94M+
INTERACTIONS

20%
SAMPLE

15
GENRES

~0.07%
MATRIX DENSITY

COSINE
SIMILARITY

UB-CF
ENGINE 1

IB-CF
ENGINE 2

01 PROJECT INFRASTRUCTURE & SCALE

- Music Info.csv:** Track ID, name, artist, genre, audio features.
- Listening History.csv:** User ID, track ID, playcount interactions.
- Merge:** Inner join on track_id; NaN rows dropped for integrity.
- Feature Removal:** Dropped spotify_preview_url, tags, time_signature.
- Sampling:** 20% sample validated against full genre distribution.
- Sparse Matrix:** Rows = user_ids · Cols = track_ids · Values = playcounts.

02 BUSINESS QUESTIONS

BQ1

What is the user's next song? Based on listening taste twins (UB-CF).

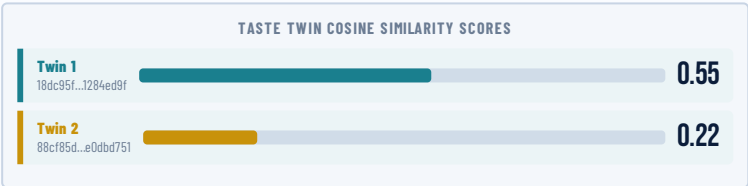
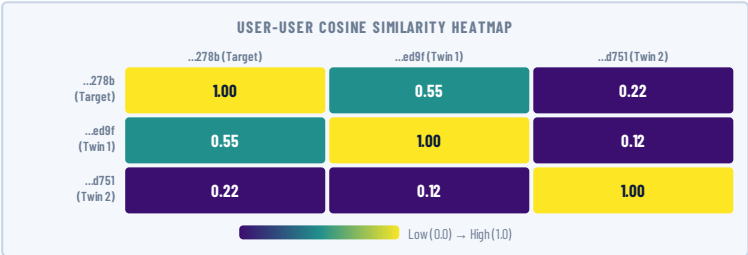
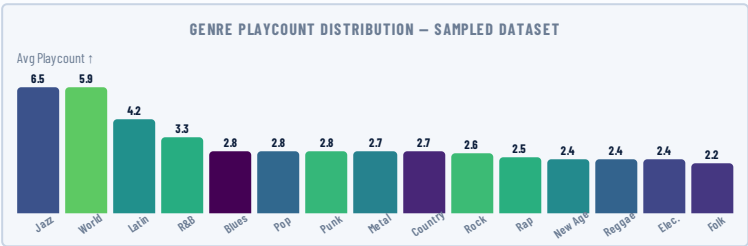
BQ2

If a user enjoys Track X, what similar tracks might they enjoy? (IB-CF).

03 METHODOLOGY

- Build User-Item Matrix** — sparse matrix from playcounts
- Cosine Similarity** — quantify user or item preference overlap
- UB-CF:** Identify taste twins; recommend unheard tracks from their history
- IB-CF:** Find items similar to seed track; rank by cosine score
- Heatmap Validation** — visualise user-user similarity matrix
- Compare outputs** — cross-validate both engines against genre signal

04 VISUALISATIONS



05 ANALYSIS — U-B-F

▶ TARGET USER PROFILE

- User ID:** 0007c0e7...8026a278b
- Era:** Contemporary & Classic Rock (1975–2020)
- Dominant Genre:** Rock — 6 of 7 recommendations
- Secondary:** Electronic — M83 (1 recommendation)
- Twin 1:** Similarity 0.5547 — strong overlap
- Twin 2:** Similarity 0.2182 — moderate overlap

▶ GENRE INSIGHT

- Jazz & World** lead avg playcounts (6.5 & 5.9).
- Rock** avg 2.6 — dominant in target recommendations.
- Electronic** avg 2.4 — present via M83.
- Sparse density ~0.07% limits lower-volume genres.

06 U-B-F TRACK RECOMMENDATIONS

Target user: 0007c0e7...8026a278b

#	TRACK NAME	ARTIST	GENRE
0	Runnin' Down a Dream	Tom Petty & The Heartbreakers	ROCK
1	Caught By The River	Doves	ROCK
2	Cubism Dream	Local Natives	ROCK
3	Airplanes	Local Natives	ROCK
4	In The Cold I'm Standing	M83	ELECTRONIC
5	Wide Eyes	Local Natives	ROCK
6	Shanty for the Arethusa	The Decemberists	ROCK

07 I-B-F TRACK RECOMMENDATIONS

SEED TRACK
Something So Strong
Crowded House · ROCK

#	TRACK NAME	ARTIST	GENRE
0	Runnin' Down a Dream	Tom Petty & The Heartbreakers	ROCK
1	In The Cold I'm Standing	M83	ELECTRONIC
2	Caught By The River	Doves	ROCK
3	Cubism Dream	Local Natives	ROCK
4	Wide Eyes	Local Natives	ROCK
5	Shanty for the Arethusa	The Decemberists	ROCK
6	Airplanes	Local Natives	ROCK

▶ KEY OBSERVATIONS

- Rock dominant:** 6 of 7 tracks match seed track genre.
- Local Natives ×3:** High item-item similarity to seed.
- M83 outlier:** Electronic — stylistic crossover detected.
- Genre signal:** IB-CF effectively captures genre preference.

08 FINDINGS

- Robust Pipeline:** Metadata and listening history merged and cleaned.
- Representative Sample:** 20% mirrors full dataset genre distribution.
- Dual Engine:** Both UB-CF and IB-CF generated 7 personalised tracks.
- Stable Similarity:** Cosine scores 0.55 & 0.22 confirm measurable overlap.
- Genre Consistency:** Rock dominated both engines (6/7 tracks each).
- Engine Agreement:** UB-CF and IB-CF returned identical track sets — strong signal convergence.

09 LIMITATIONS

- Cold Start:** No recommendations for new users or unplayed tracks.
- Sparsity:** ~0.07% density weakens overlap signals.
- Popularity Bias:** High-playcount tracks dominate outputs.
- Implicit Feedback:** Playcount ≠ preference.
- Genre Bias:** Rock concentration skews output diversity.
- Scalability:** Cosine similarity is computationally intensive at scale.

10 FUTURE DIRECTIONS

- Hybrid Filtering:** Combine UB-CF + IB-CF + content-based to address cold start.
- Matrix Factorisation:** SVD / ALS for latent feature discovery in sparse data.
- Alternative Metrics:** Pearson correlation or Jaccard for richer matching.
- Diversity Re-ranking:** Promote genre variety in recommendation outputs.
- Explicit Feedback:** Supplement playcounts with ratings, likes, skips.
- Scalability:** Apache Spark for distributed matrix operations.
- Evaluation:** A/B testing — precision, recall, novelty & diversity metrics.